HeatWise AI: Role-Specific Heatwave Advisory System

A Generative AI Decision-Support Framework for Heat Risk Communication

GitHub Repository: github.com/mitm2006/heat_forcast

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Abstract

Heatwave bulletins issued by meteorological agencies describe the physical state of the atmosphere in terms of maximum temperature, departure from normal and categorical alert colours. These quantities are accurate but are rarely sufficient for the people who must act on them: a school principal, a construction supervisor, a caregiver of an elderly relative, an event organiser and a municipal officer each need a different set of decisions from the same forecast. This report presents HeatWise AI, a role-specific heatwave advisory system whose central component is a generative language model constrained by verified meteorological facts. A deterministic climatological pipeline produces a forecast, a climatological departure, a categorical risk tier aligned with India Meteorological Department (IMD) criteria and a composite risk score. These verified values are injected as immutable context into a grounded prompt for the LLaMA-3.3-70B model served through Groq, which returns a structured advisory containing a summary, a priority alert, precautions, a time-bound checklist and an emergency protocol, written in English, Hindi or Marathi for one of five stakeholder roles. A pre-compiled matrix of sixty authoritative advisories derived from National Disaster Management Authority (NDMA) and IMD guidance guarantees continuity of service whenever generative inference is unavailable. The report describes the architecture, the meteorological foundation, the grounding and guardrail design of the generative engine, the delivery interface and an operational output for school administrators in Mumbai.

1 Introduction

1.1 Motivation

Extreme heat is among the most consequential weather hazards affecting the Indian subcontinent. Its effects are unevenly distributed across society: outdoor labourers experience occupational heat stress, children in schools and school transport are exposed during midday hours, older adults with cardiovascular or renal conditions are physiologically less able to compensate for heat, large public gatherings concentrate thousands of people in open spaces, and municipal governments must coordinate water supply, cooling shelters and hospital readiness across entire cities.

Official warning products communicate hazard severity through temperature thresholds, departures from climatological normals and colour-coded alerts. While these products are essential, they leave a translation step to the reader: converting a statement such as "severe heatwave, departure of more than six degrees" into a concrete schedule, a set of precautions and an escalation plan appropriate to a particular institution. In practice this translation is performed inconsistently, in a single language, and often too late.

1.2 Problem Statement

The problem addressed in this work is the absence of an automated, trustworthy mechanism that converts verified heatwave forecasts into actionable guidance that is (i) specific to the role of the reader, (ii) written in the language the reader uses, (iii) consistent with national disaster-management and meteorological guidance, and (iv) available even when external services fail. Free-form generative models can produce fluent and tailored text, but in a life-safety setting the risk of fabricated temperatures, invented thresholds or unsafe medical statements is unacceptable. The system must therefore combine the expressive power of generative models with strict factual control.

1.3 Objectives

The objectives of HeatWise AI are as follows.

1. Build a climatological risk pipeline that produces leakage-safe forecasts, climatological departures, IMD-aligned risk tiers and a continuous risk score.
2. Design a generative advisory engine that is strictly grounded in the verified outputs of that pipeline and cannot alter them.
3. Provide differentiated advisories for five stakeholder roles in English, Hindi and Marathi using appropriate disaster-management vocabulary.
4. Guarantee service continuity through a deterministic, authoritative fallback knowledge base.
5. Deliver advisories through an interactive interface that supports role selection, language selection, checklist tracking and geospatial risk visualisation.

1.4 Contributions

The principal contributions of this work are a grounded generation architecture that separates deterministic meteorological computation from generative communication; a structured advisory schema that makes model output directly usable by software; a role-conditioned prompt design supporting three languages; a sixty-entry authoritative fallback matrix derived from published Indian guidance; and an integrated delivery interface that exposes provenance information so that readers know whether an advisory was generated or retrieved from the verified baseline.

1.5 Organisation of the Report

Section 2 summarises the background and the frameworks on which the system is based. Section 3 presents the overall architecture. Section 4 describes the meteorological foundation that supplies verified facts to the generative engine. Section 5, the core of this report, details the HeatWise AI generative advisory engine. Section 6 describes the service interface and delivery layer. Section 7 presents an operational output for a school-administrator advisory in Mumbai. Section 8 discusses strengths, limitations and future directions, and Section 9 concludes.

2 Background and Related Frameworks

2.1 Heatwave Criteria and Institutional Guidance in India

The IMD defines heatwave conditions using absolute maximum temperature thresholds together with departures from the climatological normal, with different criteria for plains and coastal stations. The NDMA issues national guidelines for the preparation of heatwave action plans that describe preventive measures, institutional responsibilities and medical response [1]. The IMD publishes heatwave warning criteria and supports state and city Heat Action Plans [2]. For workplaces, the Occupational Safety, Health and Working Conditions Code, 2020 establishes the legal context for protecting workers from hazardous conditions [3]. HeatWise AI treats these three sources as its normative reference: the classification logic follows IMD criteria, and the precautions and emergency protocols follow NDMA and occupational-safety guidance.

2.2 From Warning to Action

A recurring observation in heat-health risk communication is that warning effectiveness depends on whether recipients understand what is expected of them. A bulletin written for the general public cannot specify that a school should curtail operating hours, that a construction contractor should adopt a split shift, or that a municipal corporation should activate cooling centres. The same forecast therefore needs to be re-expressed for each decision-maker. Producing such role-specific guidance manually for every city, day and language is not scalable, and static templates fail to reflect the actual temperature and anomaly on a given day.

2.3 Grounded Generation with Large Language Models

Large language models can produce fluent, context-sensitive text in many languages [8]. They are also known to produce statements that are not supported by their input, a phenomenon commonly described as hallucination [9]. A widely used mitigation is to condition generation on externally supplied, trusted context rather than relying on parametric knowledge, as in retrieval-augmented generation [10]. HeatWise AI applies the same principle with a structured context: instead of retrieving passages, it supplies a compact record of verified meteorological quantities and instructs the model to treat them as immutable facts. This is combined with low-temperature sampling and a schema-constrained output format, so that the model acts as a communication layer rather than a source of meteorological information.

3 System Architecture

HeatWise AI is part of a larger climate intelligence platform for heatwave monitoring, prediction and early warning. The platform is organised into four layers, shown in Figure 1. The data acquisition layer retrieves historical reanalysis and operational forecasts. The forecasting and risk layer engineers features, trains regression models, computes climatological normals and departures, and assigns IMD-aligned risk tiers. The generative advisory layer, which is the focus of this report, assembles grounded context, enforces guardrails, invokes the language model and falls back to a verified rules engine when necessary. The delivery layer exposes the results through a REST service and an interactive dashboard.

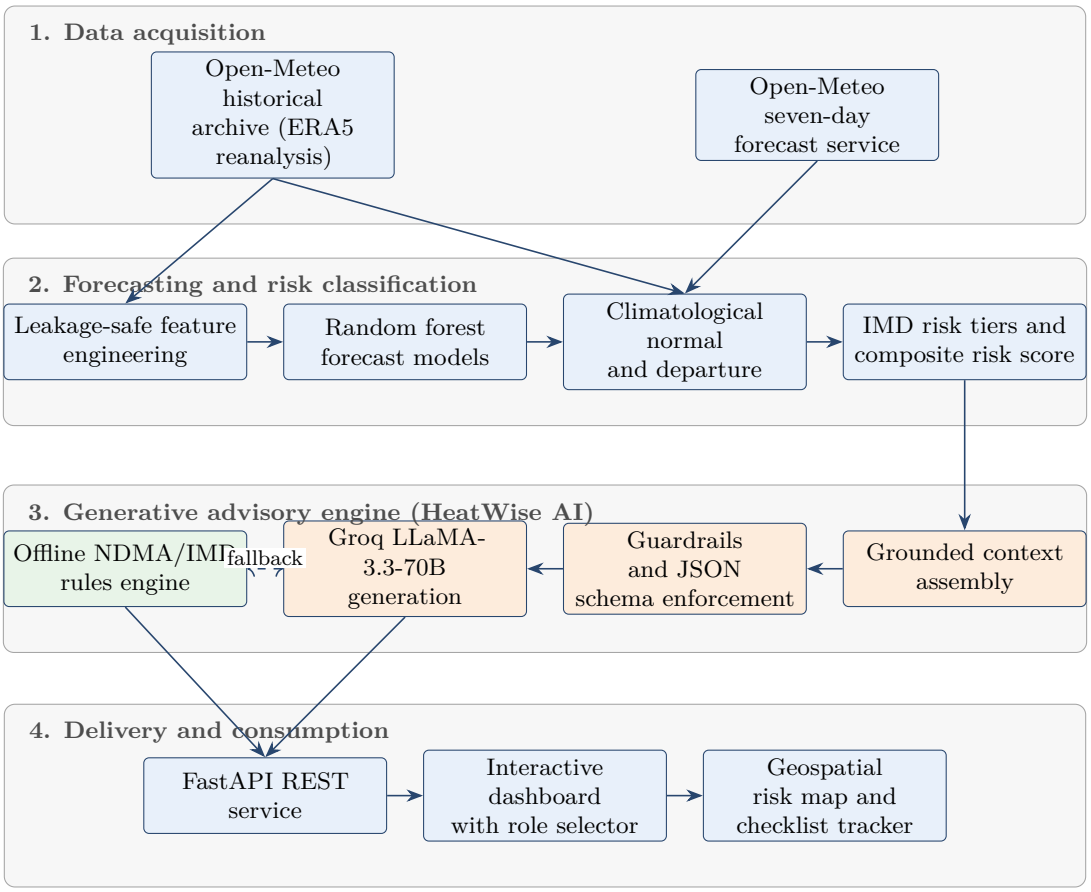


Figure 1. Layered architecture of the climate intelligence platform. The generative advisory engine receives only verified outputs of the forecasting and risk layer.

Four design principles govern the architecture.

Separation of facts and language. All meteorological quantities are computed deterministically. The generative model never produces or modifies them.

Graceful degradation. If generative inference is unavailable, the system returns an authoritative pre-compiled advisory with the same structure, so that downstream components are unaffected.

Role and language specificity. Every advisory is conditioned on a stakeholder role and an output language.

Transparency. Each response carries explicit provenance information indicating whether it was

generated by the model or retrieved from the verified baseline.

4 Meteorological Foundation

This section describes the deterministic pipeline that produces the verified facts consumed by the generative engine. Because the quality and credibility of every advisory depends on these values, the pipeline is designed to avoid temporal leakage and to follow IMD operational criteria.

4.1 Data Acquisition and Monitored Regions

Historical daily records are obtained from the Open-Meteo archive [5], which serves ERA5 reanalysis data [4], covering the period from 1 January 2015 to the present. Hourly two-metre temperature, relative humidity and precipitation are aggregated into daily maximum temperature, minimum temperature, mean temperature, humidity and rainfall. Operational seven-day forecasts are obtained from the Open-Meteo forecast service and aggregated in the same manner.

The system monitors five metropolitan regions that represent different climatological regimes (Table 1). The base threshold is the maximum temperature at which a region enters the heat-watch category.

City	**Lat.**	**Lon.**	**Type**	**Climatological regime**	**Base $T_{\max}$**
Pune	18.5204	73.8567	Plains	Semi-arid Deccan Plateau	40.0°C
Mumbai	19.0760	72.8777	Coastal	Humid tropical coastal	37.0°C
Nagpur	21.1458	79.0882	Plains	Continental central India with severe dry heat	40.0°C
Delhi	28.6139	77.2090	Plains	Subtropical semi-arid northern plains	40.0°C
Ahmedabad	23.0225	72.5714	Plains	Hot semi-arid western India	40.0°C

Table 1. Monitored metropolitan regions.

4.2 Leakage-Safe Feature Engineering

Forecast models are trained on a supervised table in which every predictor is lagged by at least one day, so that no information from the target day or later enters the feature vector. Fifteen predictors are constructed.

- **Lagged extremes (seven features):** $T_{\max}$ at lags of 1, 2, 3, 7 and 14 days, and $T_{\min}$ at lags of 1 and 7 days.
- **Rolling statistics (four features):** trailing means of $T_{\max}$ over 3, 7 and 14 days, and of $T_{\min}$ over 3 days.
- **Cyclical harmonics (four features):** sine and cosine encodings of day of year and month.

For a window length $k \in \{3, 7, 14\}$ the trailing mean is

$$\bar{T}^{(k)}_{\max}(t) = \frac{1}{k} \sum_{j=1}^{k} T_{\max}(t-j), \tag{1}$$

and the seasonal encodings are

$$\text{day_sin} = \sin\left(\frac{2\pi\,\text{DOY}}{365.25}\right), \quad \text{day_cos} = \cos\left(\frac{2\pi\,\text{DOY}}{365.25}\right), \tag{2}$$

$$\text{month_sin} = \sin\left(\frac{2\pi\,M}{12}\right), \quad \text{month_cos} = \cos\left(\frac{2\pi\,M}{12}\right). \tag{3}$$

4.3 Forecast Models

For each city, two random forest regressors [6] are fitted using scikit-learn [7]: one for $T_{\max}$ and one for $T_{\min}$. Models are fitted on a chronological 80/20 split of the record so that evaluation always concerns later dates than training. Performance is summarised by mean absolute error, root mean squared error and the coefficient of determination, with R^2 values of approximately 0.85 to 0.94 reported across seasonal variation. The fitted models are serialised for use in forecast processing, where they are applied over the seven-day horizon.

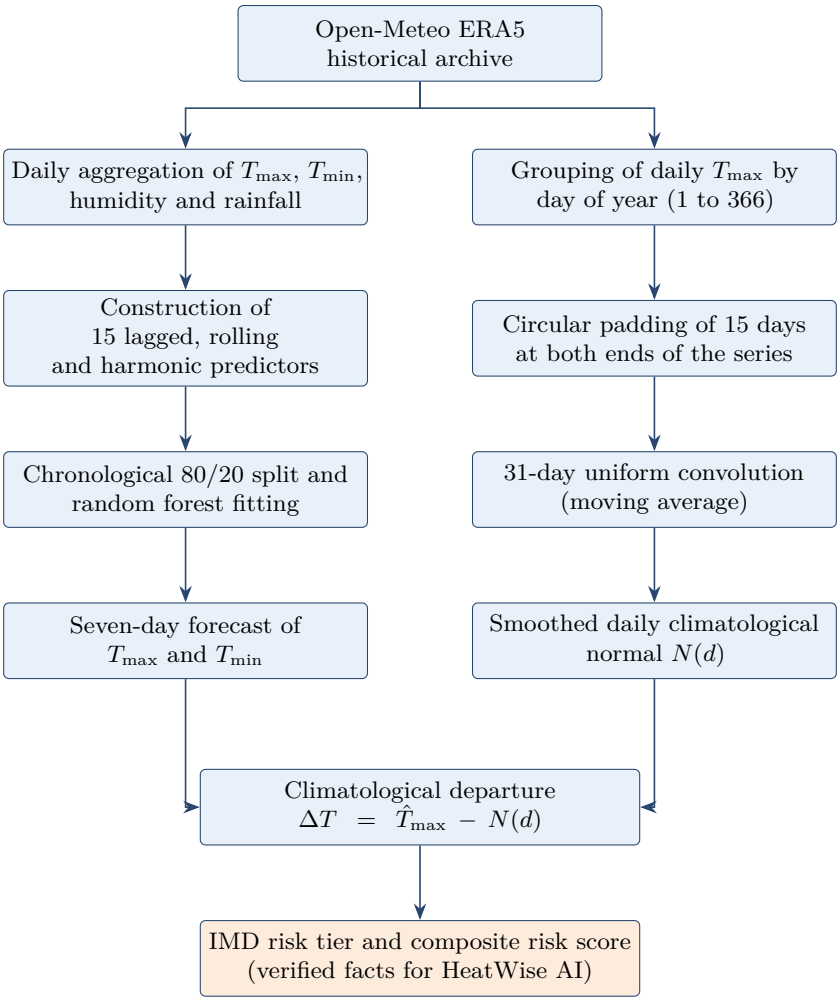


Figure 2. Forecasting and climatology paths that converge on the verified risk record.

4.4 Climatological Normal and Departure

A climatological normal is computed for every calendar day. Historical daily maxima are averaged by day of year $d \in [1, 366]$ to give a raw climatology $A(d)$. To remove the discontinuity between 31 December and 1 January, the 366-day vector is extended by 15 days at each end,

$$A_{\text{ext}} = [A_{352}, \ldots, A_{366},\, A_1, \ldots, A_{366},\, A_1, \ldots, A_{15}], \tag{4}$$

and a 31-day uniform moving average is applied,

$$N(d) = \frac{1}{31} \sum_{k=-15}^{15} A(d+k). \tag{5}$$

The result is a smooth normal curve that is continuous across the year boundary and free of localised weather anomalies. The departure for a forecast day is

$$\Delta T(t) = \hat{T}_{\max}(t) - N\big(d(t)\big). \tag{6}$$

4.5 IMD-Aligned Risk Classification

Each forecast day is assigned one of four tiers: GREEN (normal), YELLOW (heat watch), ORANGE (heatwave) and RED (severe heatwave). The criteria differ between plains and coastal stations (Table 2), reflecting the fact that a moderate absolute temperature is more significant in a humid coastal environment. The evaluation proceeds from the most severe tier downward, as shown in Figure 3.

Tier	**Plains stations (Pune, Nagpur, Delhi, Ahmedabad)**	**Coastal station (Mumbai)**
RED	$T_{\max} \geq 47.0°C$, or ($T_{\max} \geq 40.0°C$ and $\Delta T \geq 6.5°C$)	$T_{\max} \geq 45.0°C$, or ($T_{\max} \geq 37.0°C$ and $\Delta T \geq 6.5°C$)
ORANGE	$T_{\max} \geq 45.0°C$, or ($T_{\max} \geq 40.0°C$ and $\Delta T \geq 4.5°C$)	$T_{\max} \geq 37.0°C$ and $\Delta T \geq 4.5°C$
YELLOW	$T_{\max} \geq 40.0°C$, or $\Delta T \geq 4.5°C$, or ($T_{\max} \geq 37.0°C$ and humidity $\geq 75\%$)	$T_{\max} \geq 37.0°C$, or $\Delta T \geq 4.5°C$
GREEN	All other conditions	All other conditions

Table 2. Region-specific criteria for risk tier assignment.

4.6 Composite Risk Score

To provide a continuous indicator for trend display and alert triggering, an explainable three-factor score is computed from the maximum temperature, the positive departure and the humidity:

$$\text{Risk Score} = S_{\text{temp}} + S_{\text{anomaly}} + S_{\text{humidity}}, \tag{7}$$

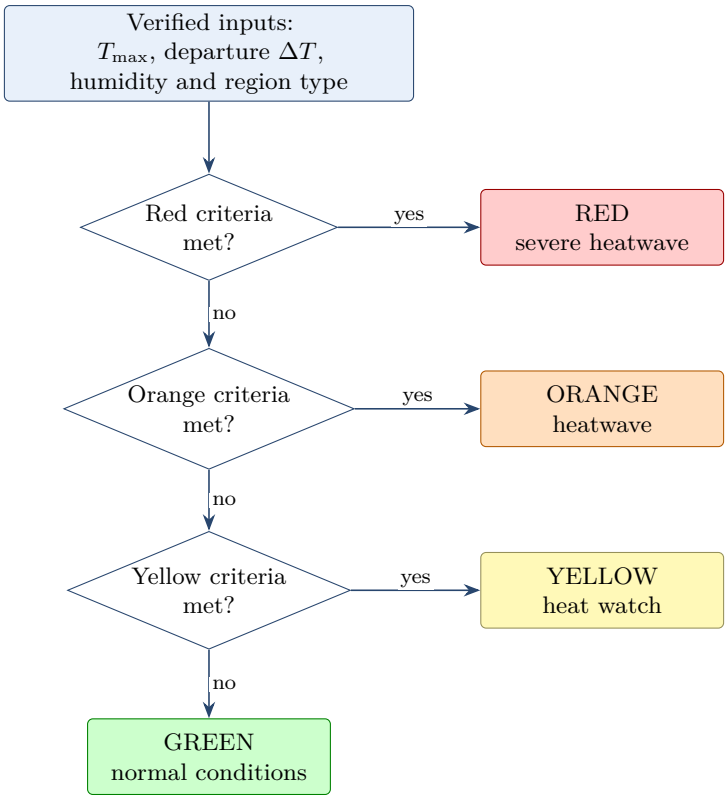


Figure 3. Order of evaluation for risk tier assignment. Criteria for each test are given in Table 2.

where

$$S_{\text{temp}} = \text{clip}\left(\frac{T_{\max} - 32.0}{15.0} \times 50.0,\ 0,\ 50\right), \tag{8}$$

$$S_{\text{anomaly}} = \text{clip}\left(\frac{\max(\Delta T,\ 0)}{7.0} \times 25.0,\ 0,\ 25\right), \tag{9}$$

$$S_{\text{humidity}} = \text{clip}\left(\frac{H - 40.0}{50.0} \times 15.0,\ 0,\ 15\right), \tag{10}$$

and H is the relative humidity in percent. The three components contribute at most 50, 25 and 15 points respectively. Because each term is individually interpretable, the score can be explained to an operator, and it is passed to the generative engine as one of the verified facts.

5 HeatWise AI: The Generative Advisory Engine

5.1 Role of Generative AI in the System

The advisory translation problem has a combinatorial character. A single verified forecast record must be expressed differently for each of five stakeholder roles, in three languages, for four alert levels, and with reference to the actual temperature and departure on the day in question. Fixed templates handle this poorly: they read generically, they cannot refer naturally to the day's specific values, and they multiply in number as roles and languages are added. A generative language model, by contrast, can integrate the specific facts with the operating context of a role and produce fluent text in each language.

The same property creates the central risk. In heatwave response, an invented temperature, a

fabricated threshold or an unsafe medical instruction can lead to injury or loss of life. HeatWise AI is therefore designed on the principle that the generative model is a *communication layer* placed on top of a deterministic meteorological core. It decides how to express and organise guidance, but it does not decide what the weather is.

5.2 Grounding Strategy

Figure 4 summarises the grounding design. Three mechanisms operate together.

1. **Parameter hardening.** The maximum temperature $T_{\max}$, minimum temperature $T_{\min}$, departure ΔT, categorical risk level (GREEN, YELLOW, ORANGE or RED) and risk score are computed by the meteorological pipeline and injected into the prompt as immutable facts. The model is explicitly prohibited from predicting, extrapolating or altering any meteorological value.
2. **Deterministic sampling.** Inference is performed at a temperature of 0.2. This suppresses creative variance and favours compliance with established public-health protocols over stylistic diversity.
3. **Structured output enforcement.** The inference request specifies a JSON object response format, and the prompt supplies the exact schema to be produced. The response can therefore be parsed programmatically without removing markdown wrappers or preamble text.

5.3 Prompt Design and Context Assembly

The prompt is composed of a system message and a user message. The system message establishes the identity of the model as the HeatWise AI heatwave advisor and states its objective: to convert verified weather alerts into personalised, practical and actionable safety advisories. It then specifies five strict rules.

1. Every recommendation must be grounded in the verified weather data provided.
2. The verified forecast values are absolute facts, and the model must not introduce different temperatures or invent weather signals.
3. All advice must be aligned with NDMA and IMD standards.
4. Every textual field must be written in natural, fluent, native-level text in the requested target language.
5. The output must be exactly one JSON object conforming to the supplied schema, without markdown wrapping.

The user message carries the verified record for the request: the city, date and region type, $T_{\max}$, $T_{\min}$, departure, risk level and risk score, together with the role descriptor and the target language. The schema supplied to the model is shown below.

Listing 1. Advisory schema supplied to the model.

```
{
  "summary": "2-3 sentences stating the exact temperature, departure and group risk",
```

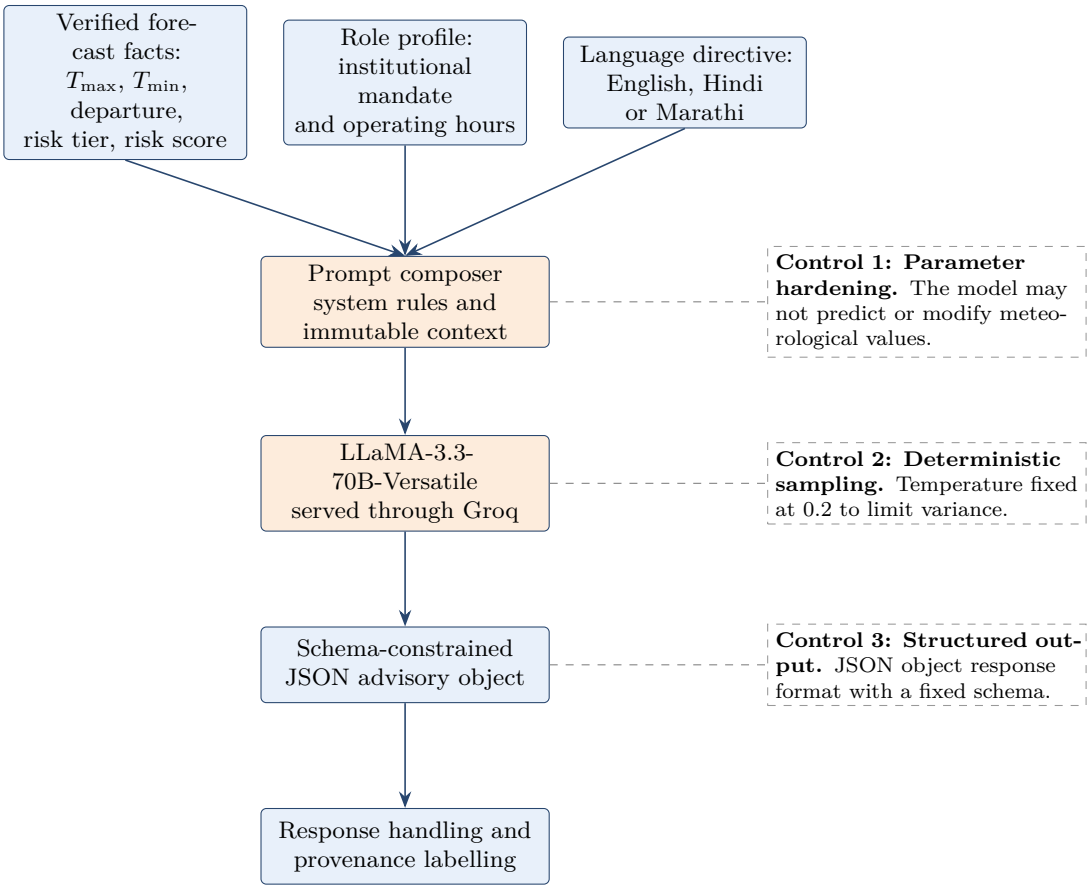


Figure 4. Grounded generation design. Deterministic inputs enter the prompt as fixed facts, and three controls constrain the behaviour of the model.

```
  "priority_alert": "One headline or imperative instruction for this role",
  "precautions": ["precaution 1", "precaution 2", "precaution 3", "precaution 4"],
  "actionable_checklist": [
    {"task": "Task description", "timeframe": "07:00 or Morning", "priority": "High|Medium"}
  ],
  "emergency_protocol": "Steps if someone suffers heat exhaustion or heat stroke",
  "official_citations": ["NDMA Heat Wave Guidelines", "IMD Heat Action Plan"]
}
```

Each field serves a distinct purpose in the interface (Table 3). Because the schema is fixed, the generative output and the fallback output share the same structure, and the interface renders both identically.

Field	Purpose
`summary`	Two to three sentences placing the exact temperature and departure in the context of the target group.
`priority_alert`	A single headline or imperative instruction shown prominently as the primary message.
`precautions`	A list of role-specific protective measures.
`actionable_checklist`	Time-bound tasks with a timeframe and a priority label, rendered as an interactive checklist.
`emergency_protocol`	Practical first-response steps for heat exhaustion or heat stroke within the group.
`official_citations`	References to the institutional frameworks that the advisory follows.

Table 3. Fields of the advisory object.

5.4 Role-Specific Personalisation

A threshold of $42°C$ implies entirely different operational responses depending on who is reading. HeatWise AI supports five roles (Table 4), each defined by a focus area and an institutional mandate that is supplied to the model as part of the role descriptor.

Identifier	Cohort	Focus and institutional mandate
`outdoor_workers`	Outdoor and construction workers	Occupational heat stress mitigation, split-shift rescheduling (06:00 to 11:00 and 16:30 to 19:30), mandatory shaded hydration breaks, buddy monitoring and wet-bulb cooling.
`school_admins`	School administrators and educators	Child protection, suspension of midday outdoor sports and assemblies (10:30 to 15:30), classroom cross-ventilation, scheduled water-break drills and cool transit dispatch.
`older_adults`	Older adults and vulnerable groups	Geriatric public health, monitoring of chronic cardiovascular and renal symptoms, mitigation of heat-induced vasodilation, passive indoor cooling and caregiver check-in schedules.
`event_organisers`	Event organisers and crowd managers	Mass gathering risk reduction, shaded staging, misting fans, free drinking-water kiosks, medical tent readiness and triage of heat exhaustion in crowds.
`municipal_officials`	Municipal officials and city planners	Activation of City Heat Action Plans, re-routing of water tankers to informal settlements, opening of public cooling centres and hospital heat-ward readiness.

Table 4. Supported stakeholder roles.

5.5 Native Multilingual Generation

HeatWise AI generates advisories in English, Hindi and Marathi. The requirement is not literal translation but fluent text that uses the vocabulary of disaster-management communication in each language, rendered in native Devanagari script in the application. Table 5 summarises the register and typical terminology; terms are transliterated here for typesetting.

Language	Code	Register and representative terminology
English	`en`	Formal, operational tone suited to administrative and enterprise execution.
Hindi	`hi`	Culturally grounded terms such as *lu* (heatwave), *ushmaghat* (heat stroke), *jal ghanti* (water bell) and *chhayadar vishram sthal* (shaded rest zones).
Marathi	`mr`	Terminology used in Maharashtra State disaster management plans, such as *ushnatechi lat* (heatwave), *tak va limbu sarbat* (electrolytic fluids) and *prabhag adhikari* (ward officers).

Table 5. Supported output languages.

5.6 Authoritative Fallback Knowledge Base

Emergency communication cannot depend on the availability of a third-party service. HeatWise AI therefore embeds a deterministic reference knowledge base that is used whenever a Groq key is absent, the service is rate-limited, a network error occurs or the inference result is unusable. The knowledge base contains one authoritative advisory for every combination of role, alert level and language:

$$5 \text{ roles} \times 4 \text{ alert levels} \times 3 \text{ languages} = 60 \text{ advisories.} \tag{11}$$

Each entry has the same structure as the generative output (Figure 5) and incorporates recommendations drawn from the NDMA national guidelines [1], the IMD Heat Action Plan and heat index criteria [2], and the Occupational Safety, Health and Working Conditions Code, 2020 [3].

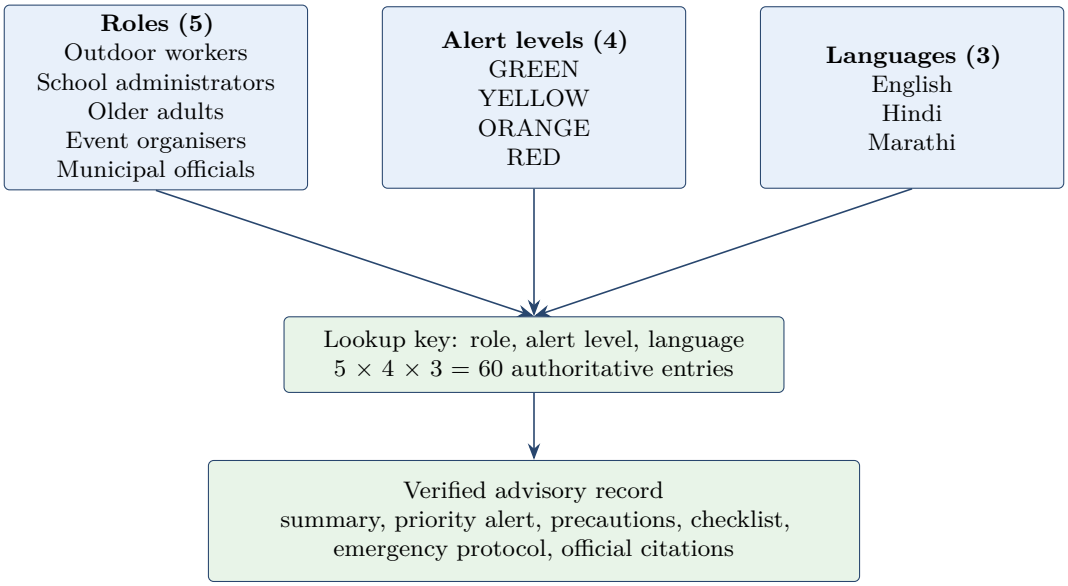


Figure 5. Structure of the sixty-entry authoritative fallback knowledge base.

The knowledge base has a second function. It is the verified baseline against which the intent of the generative output is defined: the generative engine is instructed to remain consistent with the same NDMA and IMD sources from which the baseline was compiled.

5.7 Request Lifecycle

Figure 6 shows how a request is processed. The client submits a city, role, language, date and, optionally, a user-supplied API key to the advisory endpoint of the FastAPI service. The service retrieves the verified forecast record for the city and date and passes it to the HeatWise engine. If a valid Groq key is available, the engine assembles the grounded prompt and performs inference with a temperature of 0.2 in JSON mode. If a key is absent, or if inference fails through rate limiting, a network error or an unusable response, the engine looks up the corresponding entry in the fallback matrix. In either case the service returns a structured response that includes provenance information, and the dashboard renders the precautions, checklist and emergency protocol.

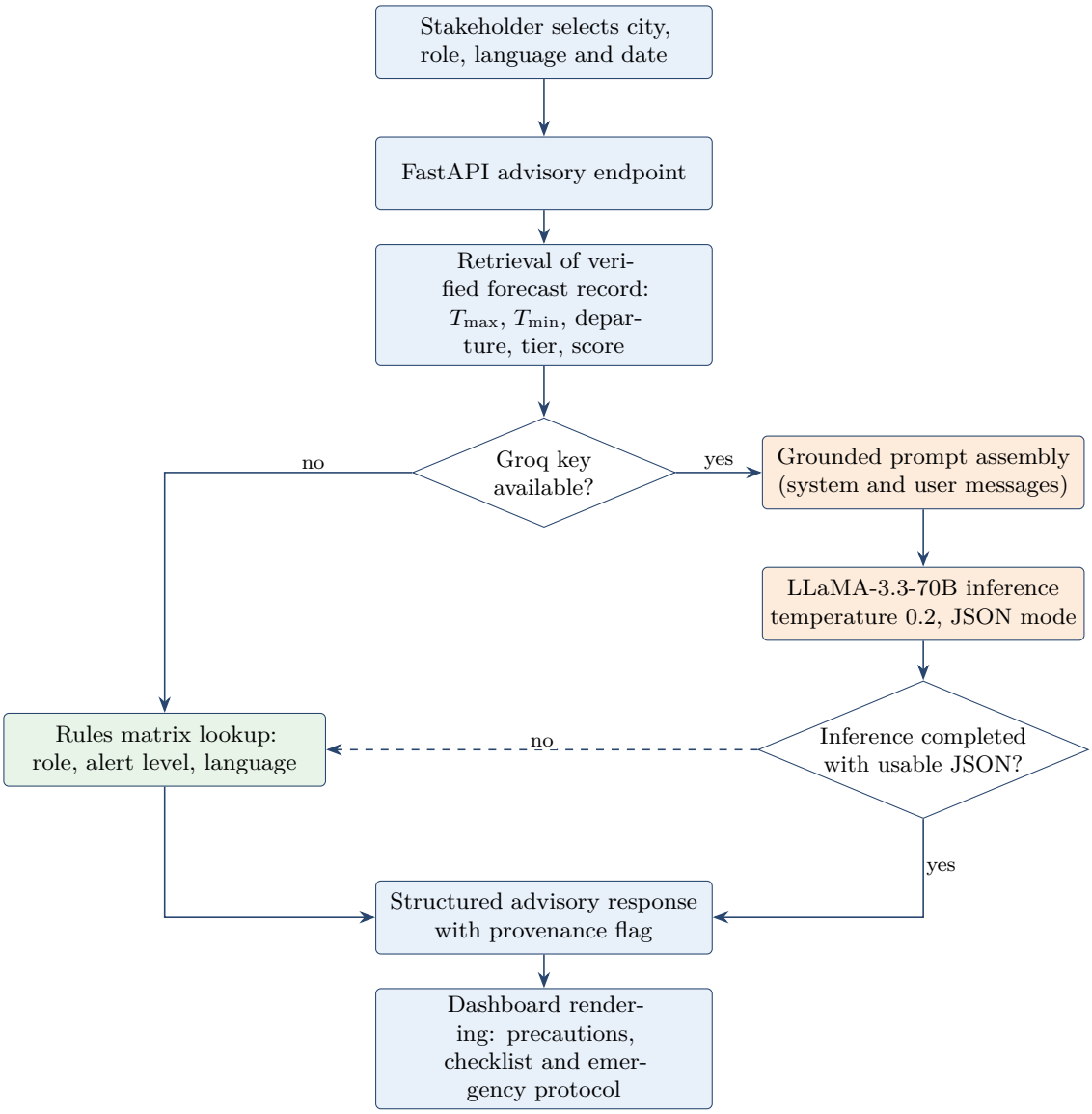


Figure 6. Lifecycle of an advisory request, showing the generative path and the deterministic fallback path.

5.8 Response Structure and Provenance

The response returned by the service contains the request context and the advisory itself (Table 6). The `is_generative` flag and the `model_used` label allow the interface and any downstream consumer to distinguish an advisory produced by the language model from one drawn from the verified baseline. This disclosure is deliberate: readers and administrators who rely on an advisory should be able to see how it was produced.

Response field	Description
`city`, `date`	Location and forecast date to which the advisory applies.
`role`, `role_label`	Role identifier and its localised display name.
`language`	Output language code.
`risk_level`, `risk_score`	Verified risk tier and composite score from the meteorological pipeline.
`model_used`	Label of the engine that produced the advisory.
`is_generative`	Boolean provenance flag: true for model output, false for the verified baseline.
`advisory`	Object conforming to the schema in Listing 1.

Table 6. Fields of the advisory response.

5.9 Safety, Accountability and Responsible Use

Several measures address the risks inherent in generative advisories for public safety.

- **Factual control.** Meteorological values are produced only by the deterministic pipeline. The model receives them as fixed inputs.
- **Normative alignment.** Both the generative prompt and the fallback matrix refer to NDMA, IMD and occupational-safety guidance, and the interface displays the framework alignment with every advisory.
- **Availability.** The fallback matrix ensures that a verified advisory can always be returned.
- **Transparency.** Provenance flags identify the source of each advisory.
- **Subordination to official orders.** Advisories are decision support. Orders issued by competent authorities, such as district collector directives for schools, remain authoritative, and advisories refer to them rather than replace them.
- **Minimal data.** A request needs only a city, a role, a language and a date. No personal data about the reader is required. Operators who wish to use generative inference supply their own Groq API key through the interface.

6 Service Interface and Delivery Layer

6.1 REST Service

The FastAPI backend exposes forecast data, model metrics and the HeatWise advisory endpoint. The advisory endpoint accepts the city, role and language, together with the forecast date and an optional user-supplied API key, and returns the structured response described in Table 6. Forecast results, risk tiers and scores are read from the processed forecast table produced by the

meteorological pipeline, so the advisory is always generated from the same verified record that the dashboard displays.

6.2 Interactive Dashboard

The web dashboard provides the HeatWise advisory centre alongside forecast cards, risk badges and an interactive OpenStreetMap layer showing risk-coloured city markers. The advisory centre is built around the following interactions.

- A role selector with five cards, one per stakeholder cohort.
- A language selector for English, Hindi and Marathi.
- A forecast window selector that lists dates together with their forecast maximum temperature and risk tier.
- A key manager through which the operator supplies a Groq API key for generative inference.
- A header summarising the verified forecast, risk score, region type and the engine that produced the advisory.
- A precaution list, an emergency protocol panel and an interactive checklist with priority labels, a preparedness progress indicator and a copy function.
- A footer listing the official frameworks to which the advisory is aligned.

Figure 7 shows the role, language and forecast-window controls of the advisory centre.

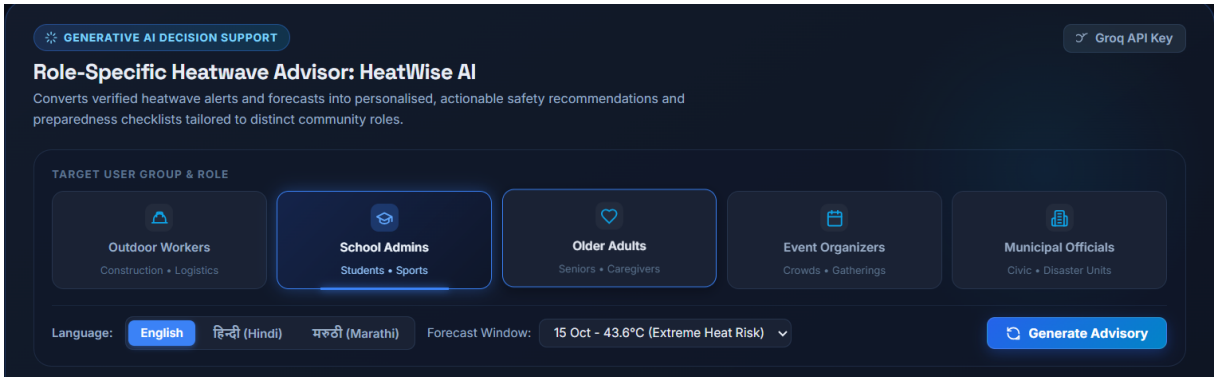


Figure 7. Advisory centre controls: role selection, language selection, forecast window and advisory generation.

6.3 Technology Stack

Table 7 lists the technologies used across the layers of the system.

Layer	Technology	Responsibility
Generative AI	Groq API, `llama-3.3-70b-versatile`	Persona-tailored advisory synthesis, multilingual generation and structured checklist generation.
Core runtime	Python 3.10 and later	Scientific computation, API services and the modelling pipeline.
Machine learning	scikit-learn random forest regressors	Regression of $T_{\max}$ and $T_{\min}$ with chronological validation.
Data processing	Pandas and NumPy	Ingestion, lag generation, circular smoothing and risk heuristics.
Backend	FastAPI and Uvicorn	Asynchronous REST API for risk metrics, forecast tables and advisories.
Frontend	Flask, Jinja2 and CSS	Responsive dark-theme web application.
Visualisation	Leaflet.js and OpenStreetMap	Geospatial risk mapping with coloured circular markers.
Deployment	Docker, Kubernetes and AWS EC2	Containerised services and cloud deployment.

Table 7. Technology stack.

6.4 Deployment

The API service and the dashboard are packaged as separate container images and orchestrated together for single-host deployment. Kubernetes manifests define deployments and services for each component and an ingress for routing, and the platform includes guidance for deployment on AWS EC2 together with tooling for right-sizing the compute resources of the workload.

6.5 Source Code Repository

The source code for HeatWise AI is available on GitHub. The repository provides the project code and related implementation files:

`https://github.com/mitm2006/heat_forcast`

7 Operational Output: School Administrator Advisory for Mumbai

This section presents an advisory produced by the system for school administrators in Mumbai for 15 October 2026, using the interface described in Section 6. It illustrates how the verified facts, the role profile and the structured schema combine into an advisory.

7.1 Verified Forecast and Risk Classification

The advisory header (Figure 8) reports the grounded verified forecast: a maximum temperature of 43.60°C, a departure of $+12.43°C$ from the climatological normal, a risk score of 67 out of 100 and the region type, coastal. Under the coastal criteria in Table 2, a maximum temperature above $37.0°C$ combined with a departure above $6.5°C$ satisfies the RED condition, and the

system consequently labels the day as extreme heat risk. The header also states the engine that produced the advisory, the NDMA and IMD grounded rules engine acting as the verified baseline.

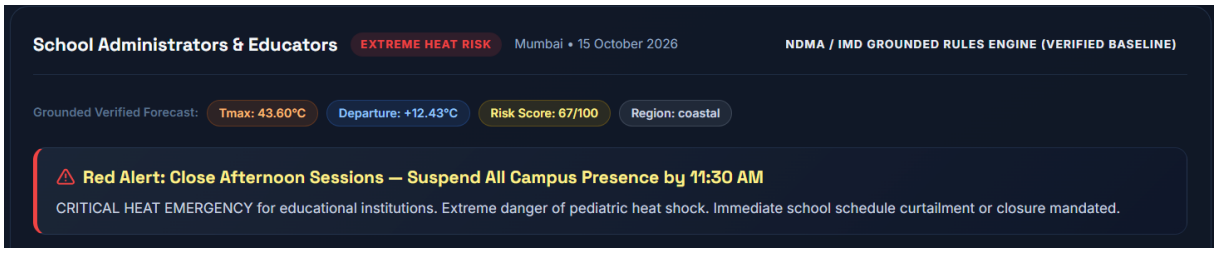


Figure 8. Advisory header for school administrators in Mumbai: verified forecast values, risk tier and priority alert.

7.2 Priority Alert

The priority alert is a single imperative headline, "Red Alert: Close Afternoon Sessions — Suspend All Campus Presence by 11:30 AM", followed by a short statement that identifies a critical heat emergency for educational institutions, the danger of paediatric heat shock and the need for immediate curtailment or closure of the school schedule. The headline addresses the specific decision that a school administrator must make, which a general bulletin does not do.

7.3 Precautions, Emergency Protocol and Checklist

Figure 9 shows the remainder of the advisory. The role-specific precautions cover the following areas.

- Curtailing operating hours to an early morning session between 07:00 and 11:00, or declaring a holiday in accordance with district collector directives.
- Prohibiting student exposure to direct sunlight and keeping children in shaded, cooled indoor environments.
- Dismissing classes before the solar peak and arranging guarded, shaded transit home.
- Strictly prohibiting children from being kept inside parked non-air-conditioned school buses.
- Coordinating with local paediatric emergency clinics for rapid heat-stroke triage.

The emergency heat-illness protocol lists the warning signs of paediatric heat shock, namely high fever, vomiting, convulsions and unconsciousness, and gives immediate steps: call the national emergency number 108, cool the child with lukewarm or cool water, fan vigorously, apply ice towels at the neck and armpits and transfer the child to intensive care.

The interactive checklist converts the guidance into time-bound tasks with priority labels (Table 8). The progress indicator records the number of completed tasks, and a copy function allows the checklist to be shared with staff.

Time	Task	Priority
06:30	Enforce the district collector order for morning-only hours, closing by 11:00 AM	High
10:30	Mobilise school buses with window shades for early, safe departure	High
11:30	Mandatory final classroom sweep to confirm that no students remain in vehicles	High
—	Station an emergency medical car on campus with chilled electrolyte saline	—

Table 8. Checklist items shown in the advisory.

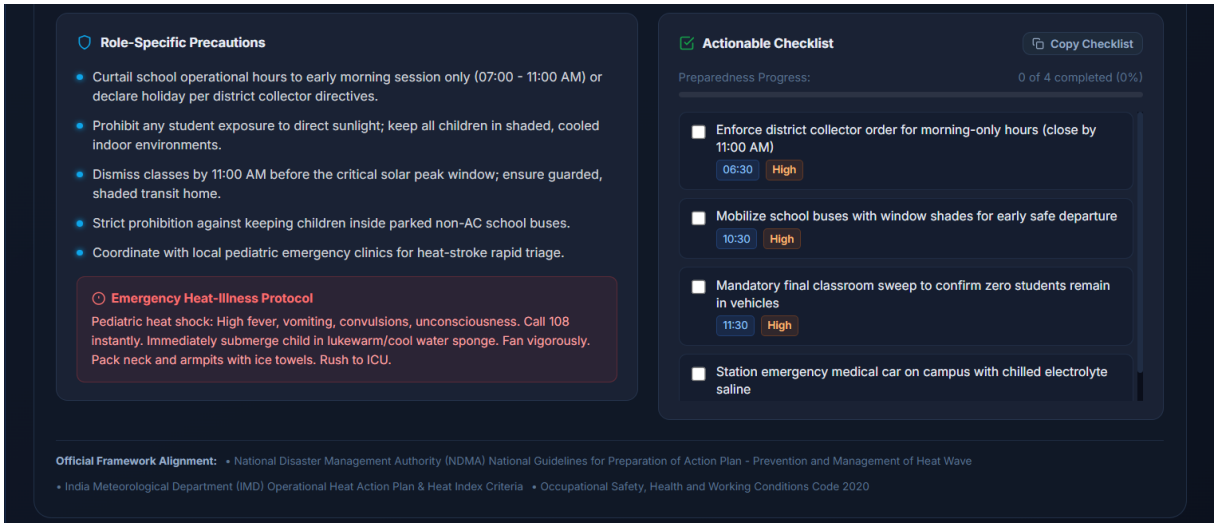


Figure 9. Role-specific precautions, emergency heat-illness protocol, interactive checklist and framework alignment.

7.4 Framework Alignment

The footer of the advisory identifies the official frameworks to which it is aligned: the NDMA National Guidelines for Preparation of Action Plan for Prevention and Management of Heat Wave, the IMD Operational Heat Action Plan and Heat Index Criteria, and the Occupational Safety, Health and Working Conditions Code, 2020. This alignment is shown with every advisory so that the reader can trace the guidance to its institutional source.

8 Discussion

8.1 Strengths

The principal strength of the system is the explicit separation between facts and language. Because the meteorological pipeline determines every quantity that appears in an advisory, the generative model can be given considerable freedom in expression without being given any freedom in substance. The structured schema makes the output machine-readable, the role profiles make it relevant, the trilingual generation makes it accessible, and the fallback matrix makes it dependable. Provenance disclosure supports trust by allowing readers to see how each advisory was produced.

8.2 Limitations

Several limitations should be recognised.

- Geographic coverage is limited to five metropolitan regions, and the classification criteria encode two region types, plains and coastal.
- Generative inference depends on an external service. The fallback matrix preserves availability, but fallback advisories are less tailored to the specific values of a given day.
- Low-temperature sampling reduces but does not eliminate variability in model output. The grounding rules and schema constrain the output, but the exact wording and ordering of recommendations can differ between requests.
- The composite risk score is a heuristic. Its weights express a judgement about the relative importance of temperature, anomaly and humidity and are not derived from health-outcome data.
- Humidity enters the risk logic through relative humidity rather than a full wet-bulb or heat-index computation.
- Three languages are supported; many communities in the monitored regions use others.

8.3 Future Directions

Several extensions follow naturally from the present design: expansion to additional cities and climate regions; support for further Indian languages; delivery through channels such as SMS, voice and messaging applications for readers without regular access to a web interface; incorporation of wet-bulb globe temperature and heat-index measures into the risk logic; structured review of advisories by public-health and disaster-management practitioners; and retrieval of local administrative orders so that advisories can cite the specific directives in force for a district.

9 Conclusion

HeatWise AI addresses the gap between a heatwave forecast and the decisions that different stakeholders must take in response. A deterministic climatological pipeline provides forecasts, departures, IMD-aligned risk tiers and an explainable risk score. A grounded generative engine based on LLaMA-3.3-70B turns these verified facts into role-specific, multilingual advisories with precautions, time-bound checklists and emergency protocols, under three controls: parameter hardening, deterministic sampling and schema-constrained output. A sixty-entry knowledge base derived from NDMA and IMD guidance guarantees that an authoritative advisory is always available, and provenance flags make the origin of each advisory visible. The result is a system that uses generative AI for what it does well, adapting language and structure to the reader, while keeping the facts on which lives depend under deterministic control.

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